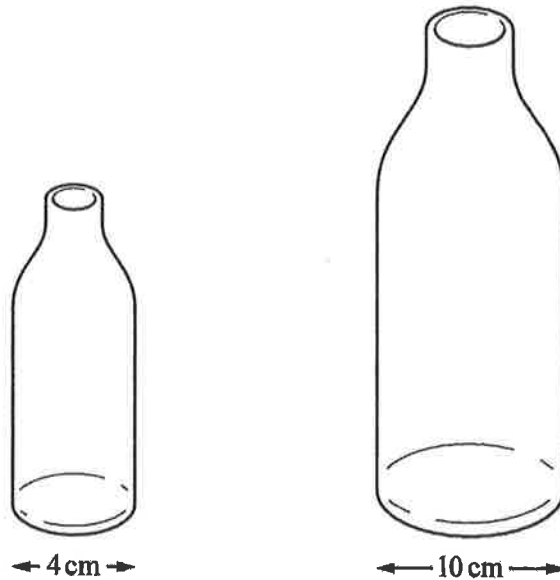


Unit 2.2 Congruence and Similarity

Paper 1: Basic

1.



The diagram shows two geometrically similar bottles.

The diameter of the base of the smaller bottle is 4 cm.

The diameter of the base of the larger bottle is 10 cm.

- (a) The height of the larger bottle is 30 cm.

Calculate the height of the smaller bottle.

[2]

- (b) The smaller bottle has a capacity of 125 ml.

The manufacturer makes another geometrically similar bottle that holds 500 ml.

Calculate the diameter of the base of this bottle.

[2]

[N19/I/20]

2. The cargo volume of an aircraft is 184 m^3 .

A model of this aircraft is built to a scale of 1 : 20.

Find the cargo volume, in m^3 , of the model.

Give your answer in standard form.

[3]

[N18/I/11]

3.



A company manufactures three sizes of the same brand of shampoo.

The bottles are all geometrically similar.

The height of the 250 ml bottle is 18.4 cm.

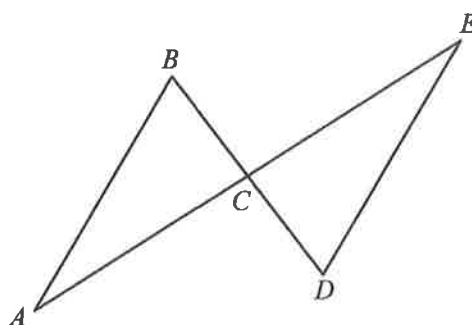
Calculate the height of the 75 ml bottle.

[3]

[N17/I/23(b)]

Paper 1: Intermediate

4.



In the diagram, AB and DE are equal in length and parallel.

ACE and BCD are straight lines.

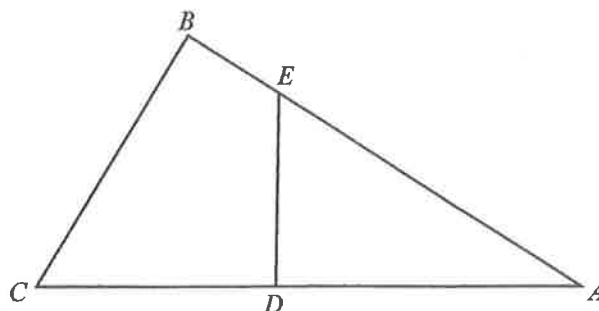
Show that BD bisects AE .

Give a reason for each statement you make.

[4]

[N22/I/15]

5.



In the diagram, ABC and ADE are triangles such that $\angle ABC = \angle ADE$.

(a) Show that the two triangles are similar.

[2]

(b) Given that $AC = 12$ cm, $AB = 10$ cm and $AE = 9$ cm, show that $CD = 4.5$ cm.

Show all your working.

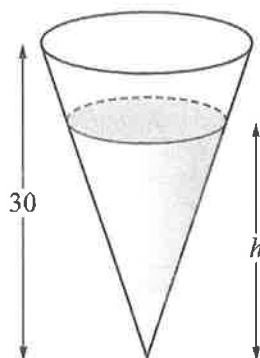
[2]

[N21/I/14]

6. (a) A cruise ship has an average fuel consumption of 0.000 925 kilometres per litre.
Write this consumption in litres per kilometre. [1]
- (b) Another cruise ship has a length of 330 m.
The capacity of the fuel tanks is 1.228×10^6 litres.
A model of the ship is made to a scale of 1 : 60.
- (i) Find the length of the model. [1]
- (ii) Find the capacity of the fuel tanks of the model. [2]

[N21/I/18]

7. The diagram shows a cone of height 30 cm.



The volume of the liquid in the cone is half the volume of the cone.

Calculate the depth, h centimetres, of the liquid.

[2]

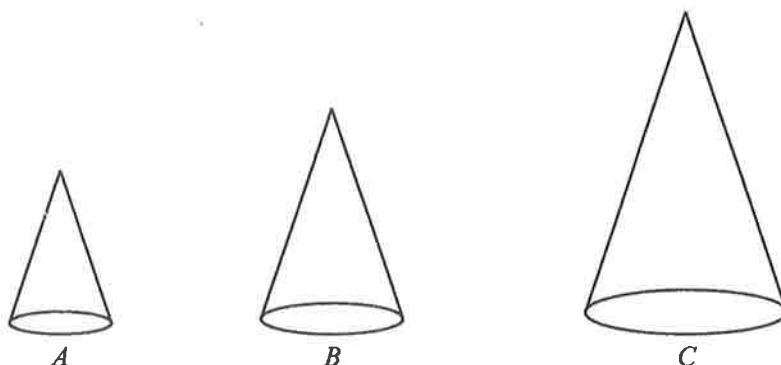
[N16/I/7]

8. Two bottles of water are geometrically similar.
The larger bottle holds 2 litres and the smaller bottle holds 1.25 litres.
The height of the larger bottle is 33.5 cm.
- (a) Calculate the height of the smaller bottle. [2]
- (b) The ratio surface area of larger bottle : surface area of smaller bottle can be written in the form $k : 1$.
Find the value of k .
Give your answer to 2 decimal places. [2]

[N15/I/17]

Paper 2: Intermediate

1.



A , B and C are similar cones.

The ratio volume of cone A : volume of cone B : volume of cone $C = 1 : 3 : 8$.

- (i) Find the ratio height of cone A : height of cone C . [1]
- (ii) Find the surface area of cone B as a percentage of the surface area of cone C . [2]

[N22/II/5(b)]

